

DATA ANALYST PORTFOLIO PROJECT

Indian Urban Air Quality Analysis

Exploratory analysis and data-integrity assessment of 10,000 AQI observations across 24 Indian cities

10,000 observations	24 cities	273.7 mean AQI	66.9% Poor or worse (AQI > 200)
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Dataset: indian_aqi_health_impact_2019_2024.csv

Tools: Python, pandas, NumPy, Matplotlib, Seaborn, SciPy, scikit-learn

Prepared by: NS | September 2026

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2. Executive Summary

Objective. Analyse air-quality readings for 24 Indian cities to identify which cities and which pollutant, weather and activity variables are associated with the Air Quality Index (AQI), and to test whether the dataset can support health-impact and trend analysis.

Dataset. 10,000 observations and 16 columns (AQI, six pollutants, five weather variables, vehicle count, industrial activity index, health impact score). There are no missing values, no duplicate rows and no AQI values outside 0–500.

273.7 mean AQI (median 273)	21.6% Severe (AQI 401–500)	0.1% Good (10 of 10,000 rows)	25.0 pts gap between highest and lowest city mean
$r \leq 0.018$ strongest correlation with AQI	0 of 13 variables significantly related to AQI	$R^2 < 0$ cross-validated model fit	1 distinct Health Impact Score value

Key analytical findings

- Air quality in the sample is poor overall.** 66.9% of observations (6,694) fall in Poor, Very Poor or Severe; only 10 (0.1%) are Good.
- Cities are statistically indistinguishable.** City means range from Ahmedabad (288.8) to Ludhiana (263.8), a 9.1% spread around the overall mean. ANOVA gives $p = 0.405$, and no city differs significantly from the rest after correcting for 24 comparisons.
- No variable explains AQI.** The strongest correlation is CO at $r = 0.018$; none of the 13 variables is significant at the 5% level, and a random-forest model scores a cross-validated R^2 of -0.004 (worse than predicting the mean).
- The data behaves like independent, evenly spread random values.** All 14 variable distributions are consistent with a flat distribution (chi-square p between 0.12 and 0.87), there are 0 IQR outliers, and pollutant readings do not rise with AQI category.
- Internal consistency is at chance level.** Under the Indian AQI method the reported category should never be lower than the category implied by the recorded pollutants. It is lower in 3,989 rows (39.9%), and the share of rows where it is at or above the implied category (60.1%) is the same as for randomly shuffled labels (60.3%).
- Two planned analyses are not possible.** Health Impact Score is constant (10 in all rows) and there is no date column, so neither health-impact nor time-trend analysis can be performed.

Major anomalies

2,160 rows (21.6%) are labelled Severe, yet no recorded pollutant reading is high enough to imply more than Very Poor. A further 1,984 rows (19.8%) carry a category at least two levels below what their own pollutant readings imply. Variable ranges also sit at round-number limits (for example PM_{2.5} spans 10.0–250.0 and PM₁₀ spans 20.0–300.0).

Real-world implication and analyst takeaway

The file is structurally clean but does not contain the relationships one expects from real monitoring data. City rankings, traffic effects and industrial effects **cannot** be inferred from it. Its value is as a demonstration of a disciplined workflow: the analysis tests each apparent pattern (for example “Ahmedabad is the most polluted city”) and shows that it does not survive statistical scrutiny. Reporting a validated negative result is the responsible conclusion here.

3. Business Problem / Objective

Urban air quality is a public-health and policy issue: authorities need to know where pollution is worst, what drives it, and whether interventions (traffic limits, industrial controls) are likely to help. This project asks whether the supplied dataset can answer those questions.

1. How severe is air quality across the sample, and how are readings distributed across AQI categories?
2. Which cities have the highest and lowest AQI, and are the differences statistically meaningful?
3. Which pollutants, weather variables, traffic and industrial measures are associated with AQI?
4. Can the dataset support health-impact modelling or a 2019–2024 trend analysis?
5. Is the data internally consistent enough to support decisions?

Analytical approach: every claim is backed by an exact value from the dataset and, where relevant, a statistical test. Where a requested analysis is impossible, the report says so explicitly.

4. Dataset Overview

The file contains **10,000 rows and 16 columns** (12 float, 3 integer, 1 text). The 24 cities have between 368 and 492 observations each (Nagpur has the most, Surat the fewest), so the sample is reasonably balanced.

Variable	Type	Role	Min	Max	Mean
City	text	Identifier	24 unique	–	–
AQI	integer	Target	50	499	273.70
PM2.5	float	Pollutant	10.02	250.00	130.41
PM10	float	Pollutant	20.02	299.98	161.00
NO2	float	Pollutant	5.01	79.99	42.45
CO	float	Pollutant	0.10	10.00	5.05
SO2	float	Pollutant	2.00	49.99	26.19
O3	float	Pollutant	5.02	100.00	52.49
Temperature (°C)	float	Weather	10.01	45.00	27.49
Humidity (%)	float	Weather	20.01	99.98	59.82
Wind speed (km/h)	float	Weather	0.50	20.00	10.23
Rainfall (mm)	float	Weather	0.03	299.98	150.61
Pressure (hPa)	float	Weather	950.00	1,050	1,000
Vehicle count	integer	Activity	1,129	499,974	249,007

Variable	Type	Role	Min	Max	Mean
Industrial activity index	float	Activity	0.00	10.00	5.00
Health impact score	integer	Outcome (constant)	10	10	10

- **Data quality:** 0 missing values, 0 duplicate rows, 0 AQI values outside 0–500, 0 IQR outliers across all numeric variables.
- **Missing information:** the file name refers to 2019–2024, but there is **no date, year or month column**. Measurement units for pollutants are not documented. The sampling method (stations, frequency) is not documented.
- **Constant column:** Health Impact Score equals 10 in all 10,000 rows and carries no information.

5. Tools & Technologies

Tool	Purpose in this project
Python 3 (Jupyter / VS Code)	Reproducible end-to-end workflow
pandas, NumPy	Loading, cleaning, aggregation, category creation, feature summaries
Matplotlib, Seaborn	13 portfolio-grade charts saved at 300 dpi
SciPy	Chi-square, ANOVA, Kruskal–Wallis, Welch t-tests, Pearson and Spearman correlation
scikit-learn	Cross-validated linear regression and random forest, used as a check on whether any predictor combination explains AQI
results.json	All report numbers are read from this file produced by the script, so text and data cannot drift apart

6. Data Cleaning & Preparation

Step	Action	Result
1. Load	Read CSV with pandas	10,000 rows × 16 columns
2. Missing values	isnull().sum() on every column	0 missing
3. Duplicates	duplicated().sum()	0 duplicates
4. Column names	Lower-case, underscores; remove brackets, %, °, . and /	e.g. "Wind Speed (km/h)" → wind_speed_kmh
5. Data types	pd.to_numeric(errors="coerce") on all numeric columns	0 values lost to coercion
6. AQI validation	Flag AQI < 0 or > 500	0 invalid records
7. AQI categories	pd.cut with bands below	6 categories created
8. Constant columns	Detect columns with one unique value	health_impact_score excluded from correlation/modelling
9. Export	Cleaned dataset and summary tables saved to CSV	aqi_analysis_output/

AQI category	Band	Observations	Share
Good	0–50	10	0.10%
Satisfactory	51–100	1,134	11.34%
Moderate	101–200	2,162	21.62%
Poor	201–300	2,312	23.12%
Very Poor	301–400	2,222	22.22%
Severe	401–500	2,160	21.60%

- **Assumption:** the category bands follow the Indian National AQI scheme (0–50, 51–100, 101–200, 201–300, 301–400, 401–500). The dataset does not document which method was used.
- **Edge case:** all 10 “Good” observations have AQI = 50 exactly, the minimum value in the file.
- **Defect found and fixed:** the initial notebook removed brackets and spaces from column names but left the “/” in “wind_speed_kmh”. Later look-ups for “wind_speed_kmh” therefore failed silently and wind speed was dropped from the environmental-factor analysis. The workflow now strips “/”; wind speed shows $r = +0.005$ with AQI.

7. Exploratory Data Analysis

7.1 AQI summary statistics

Statistic	Value	Statistic	Value
Mean	273.70	Minimum	50
Median	273	Maximum	499
Standard deviation	129.04	25th percentile	164
Skewness	0.003	75th percentile	385
Excess kurtosis	-1.18	Observations with AQI ≥ 300	4,403 (44.0%)

Skewness is essentially zero (0.003) and excess kurtosis is -1.18, almost exactly the value of a perfectly flat distribution (-1.2).

7.2 Variable distributions and uniformity

A 20-bin chi-square goodness-of-fit test compares each variable with a perfectly flat distribution. A small p-value would indicate a departure from flatness.

Variable	Min	Max	Mean	Median	IQR outliers	Uniform p
AQI	50	499	273.70	273	0	0.14
PM2.5	10.02	250.00	130.41	130.31	0	0.14
PM10	20.02	299.98	161.00	161.10	0	0.27
NO2	5.01	79.99	42.45	42.89	0	0.87
CO	0.10	10.00	5.05	5.02	0	0.12
SO2	2.00	49.99	26.19	26.44	0	0.27
O3	5.02	100.00	52.49	52.38	0	0.58

Variable	Min	Max	Mean	Median	IQR outliers	Uniform p
Temperature (°C)	10.01	45.00	27.49	27.48	0	0.35
Humidity (%)	20.01	99.98	59.82	59.89	0	0.40
Wind speed (km/h)	0.50	20.00	10.23	10.25	0	0.86
Rainfall (mm)	0.03	299.98	150.61	149.85	0	0.86
Pressure (hPa)	950.00	1,050	1,000	1,001	0	0.82
Vehicle count	1,129	499,974	249,007	247,375	0	0.67
Industrial activity index	0.00	10.00	5.00	5.02	0	0.87

None of the 14 variables departs from flatness at the 5% level, and there are no outliers under the $1.5 \times \text{IQR}$ rule.

7.3 City-level summary

Rank	City	Obs.	Mean AQI	95% CI	% Very Poor + Severe
1	Ahmedabad	435	288.8	276.8 – 300.7	48.5%
2	Kolkata	417	283.7	271.3 – 296.1	46.5%
3	Thane	433	283.4	271.6 – 295.3	45.7%
4	Jaipur	398	281.6	269.0 – 294.2	49.5%
5	Vadodara	417	279.8	267.1 – 292.4	47.0%
6	Lucknow	442	277.4	265.4 – 289.4	44.6%
7	Nashik	448	275.7	263.3 – 288.2	44.2%
8	Surat	368	275.5	262.6 – 288.3	44.0%
9	Hyderabad	418	274.6	262.5 – 286.6	44.5%
10	Chennai	419	274.2	261.8 – 286.6	43.9%
11	Rajkot	428	273.9	261.4 – 286.5	45.6%
12	Nagpur	492	273.8	262.4 – 285.2	44.3%
13	Indore	399	273.7	260.9 – 286.5	42.9%
14	Bangalore	419	272.6	260.1 – 285.1	44.2%
15	Varanasi	430	271.0	258.8 – 283.2	40.5%
16	Agra	408	270.5	257.9 – 283.1	41.9%
17	Pune	435	269.9	257.8 – 282.0	42.1%
18	Mumbai	377	269.6	256.7 – 282.6	41.1%
19	Delhi	377	269.0	255.8 – 282.3	41.9%
20	Patna	373	267.2	254.0 – 280.3	41.0%
21	Meerut	418	266.9	254.4 – 279.3	43.1%
22	Bhopal	398	265.7	252.8 – 278.6	44.2%
23	Srinagar	398	264.4	251.8 – 276.9	38.9%
24	Ludhiana	453	263.8	251.8 – 275.7	40.8%

7.4 Statistical testing summary

Test	Result	Reading
Chi-square uniformity (14 variables)	p from 0.12 to 0.87; 0 rejected	Consistent with flat distributions
Pearson correlation with AQI (13 variables)	max $ r = 0.018$ (CO); 0 significant at 5%	No linear association
Spearman rank correlation	max $ \rho = 0.019$	No monotonic association
5-fold cross-validated models	Linear $R^2 = -0.004$; random forest $R^2 = -0.004$	Predictors carry no usable signal
ANOVA / Kruskal–Wallis across cities	$F = 1.04$, $p = 0.405$; $H = 24.0$, $p = 0.405$	No detectable city effect
Chi-square: city $\times$ AQI category	$\chi^2 = 89.1$, $df = 92$, $p = 0.566$	Category mix does not differ by city
Welch t-test, each city vs. the rest	Ahmedabad: raw $p = 0.012$, Bonferroni-adjusted $p = 0.28$; 0 significant after correction	Ranking is within chance variation
PM2.5 across AQI categories (Kruskal–Wallis)	$p = 0.699$; Severe minus Satisfactory = 0.30, $p = 0.905$	Particulates do not vary with AQI
Consistency with pollutant-implied category*	reported below implied in 39.9% of rows; reported $\geq$ implied 60.1% vs. 60.3% for shuffled labels	No better than chance

*Uses external Indian AQI reference breakpoints for PM2.5, PM10, NO2, O3, CO and SO2 (units assumed, as the dataset states none) and the worst sub-category. This is an assumption, not information contained in the dataset. AQI can also be driven by pollutants not in the file, so a category ABOVE the implied one is not by itself an error; a category BELOW it is.

8. Key Visualizations

Thirteen charts were produced. Each answers one analytical question; the charts themselves appear with their insights in Section 9. No trend line chart is included because the dataset has no time field.

Fig.	Chart	Type	Analytical question
1	AQI distribution	Histogram + KDE	How are AQI readings spread, and is the shape realistic?
2	AQI category distribution	Bar chart	How much of the sample is Poor or worse?
3	Average AQI by city	Ranked bar (Top-N)	Which cities have the highest and lowest AQI?
4	PM2.5 vs AQI	Scatter + fit	Does PM2.5 track AQI?
5	PM10 vs AQI	Scatter + fit	Does PM10 track AQI?
6	Vehicle count vs AQI	Scatter + fit	Does traffic volume relate to AQI?
7	Industrial activity vs AQI	Scatter + fit	Does industrial activity relate to AQI?
8	Correlation heatmap	Heatmap	Are any variables related to each other?
9	Correlation with AQI	Ranked bar + chance band	Which variable is most associated with AQI, and is it distinguishable from noise?
10	Pollutant box plots	Box plots	Are there outliers or skew in pollutant readings?
11	City means with 95% CI	Interval plot	Are city differences statistically meaningful?
12	Particulates by AQI category	Bar + error bars	Do worse AQI categories carry more particulates?
13	Category mix by city	100% stacked bar	How does the composition of AQI categories differ by city?

Design standards: consistent palette (navy for data, red for reference lines, a green-to-maroon scale for AQI categories), titles that state the subject, labelled axes, value labels where useful, bar axes starting at zero, and statistics (r , R^2 , p , n) printed on every scatter plot.

9. Chart-by-Chart Insights

Chart: AQI Distribution

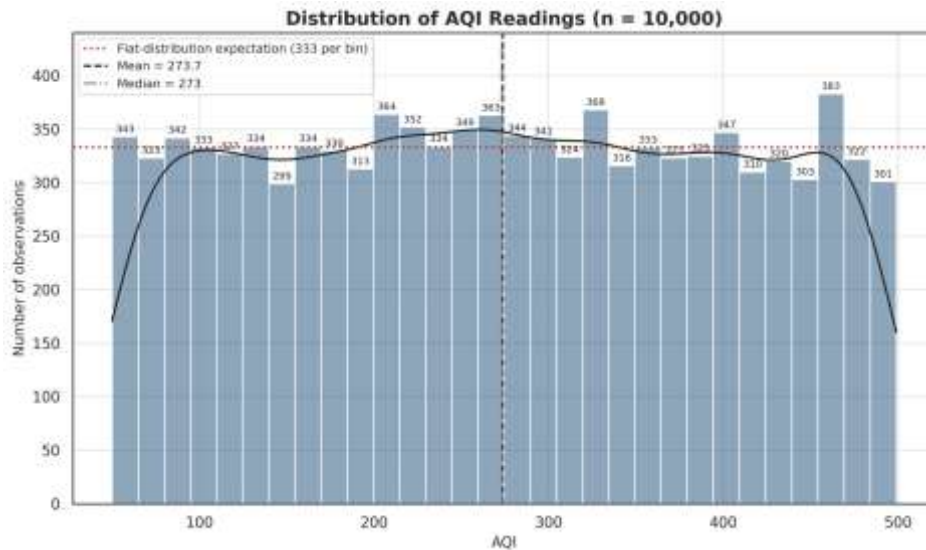


Figure 1. Distribution of AQI readings with mean, median and flat-distribution expectation

What the chart shows	Histogram of all 10,000 AQI values in 30 bins with a smoothed density curve, the mean and median, and the count each bin would contain if readings were spread evenly.
Key insight	Bin counts range only from 299 to 383 against an even-spread expectation of 333. The mean (273.7) and median (273) sit almost exactly at the midpoint of the observed range (274.5), skewness is 0.003 and excess kurtosis is -1.18. A uniformity test does not reject a flat distribution ($p = 0.14$). The only observed AQI range is 50–499; no reading falls below 50.
Business / real-world implication	Measured pollution indices normally cluster around typical levels and have a longer tail of severe episodes. A perfectly even spread from 50 to 499 means a reading of 60 is as likely as one of 270 or 490, which is unusual for real monitoring data and is the first sign that the file may not reflect real conditions.
Analyst takeaway	AQI is evenly spread across its whole range. Treat the shape as a data-plausibility flag before drawing any conclusions about “typical” air quality.

Chart: AQI Category Distribution

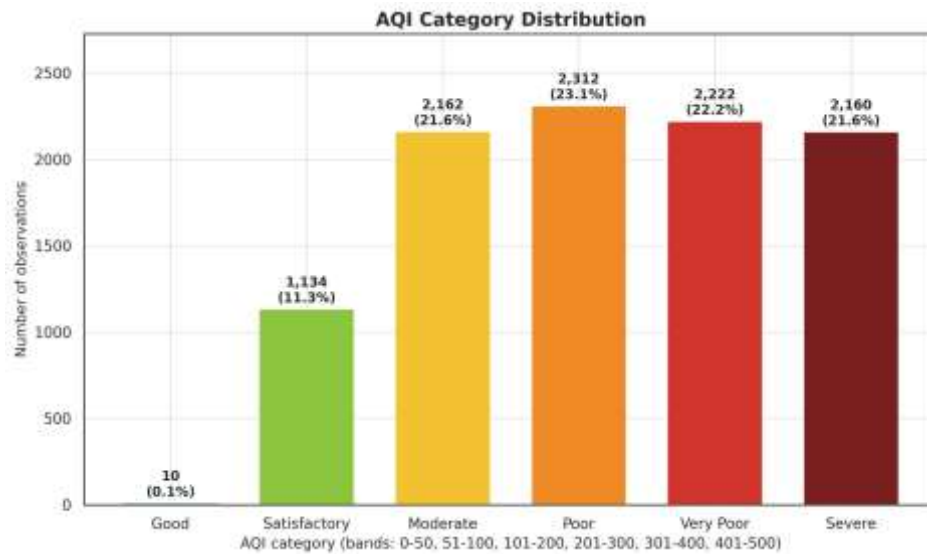


Figure 2. Number and share of observations in each AQI category

What the chart shows	Count and percentage of observations in each of the six AQI categories.
Key insight	66.9% of observations (6,694) are Poor or worse, 43.8% are Very Poor or Severe, and only 10 (0.1%) are Good. The shares track the width of each band: the four 100-point bands each hold between 21.6% and 23.1%, while the 50-point Satisfactory band holds 11.3%, about half as much.
Business / real-world implication	The headline that two-thirds of readings are Poor or worse is accurate for this sample, but because it follows mechanically from an evenly spread AQI it should not be presented as a real-world pollution profile of Indian cities.
Analyst takeaway	Category shares are a product of band width, not evidence of a pollution pattern. Cite the 66.9% figure only with this caveat.

Chart: Average AQI by City

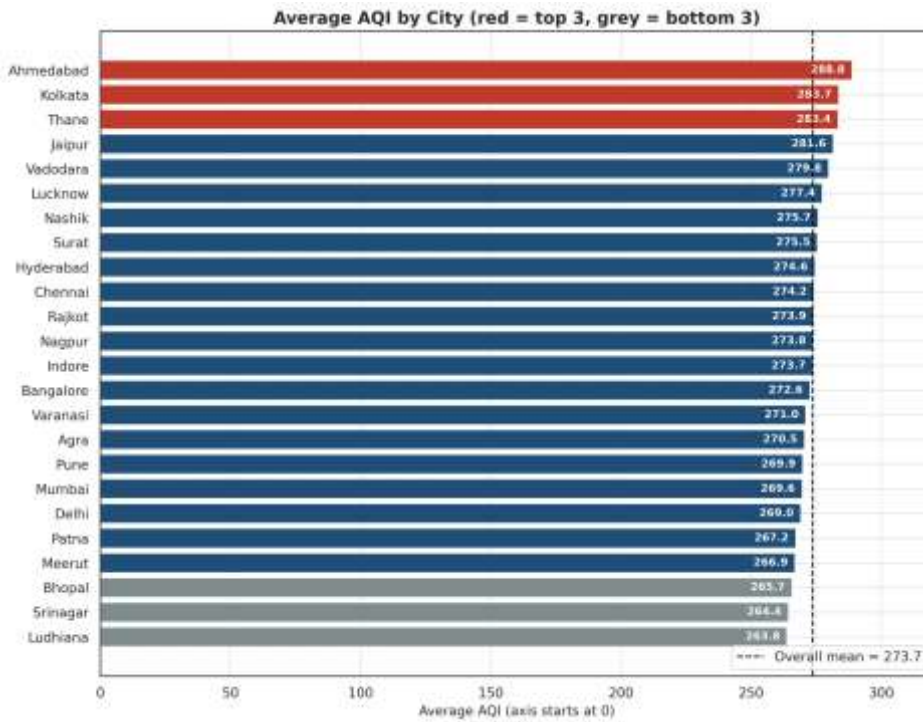


Figure 3. Mean AQI for each of the 24 cities (red = top three, grey = bottom three)

What the chart shows	Mean AQI per city, ranked, with the overall mean (273.7) as a reference line. The axis starts at zero so bar lengths are not exaggerated.
Key insight	Ahmedabad ranks first at 288.8 (15.1 above the overall mean), followed by Kolkata (283.7) and Thane (283.4). Ludhiana is lowest at 263.8 (9.9 below). The whole range is 25.0 points, 9.1% of the overall mean. Delhi ranks 19 of 24 (269.0); Surat ranks 8 (275.5).
Business / real-world implication	A ranking like this could easily be turned into a “most polluted city” headline or a priority list. The bars are almost the same length, and Figure 11 shows the gaps are within sampling noise. Well-known pollution hotspots also do not stand out here, which is a further plausibility flag.
Analyst takeaway	Ranking exists, but the 25.0-point spread is small. Do not prioritise cities on this chart alone.

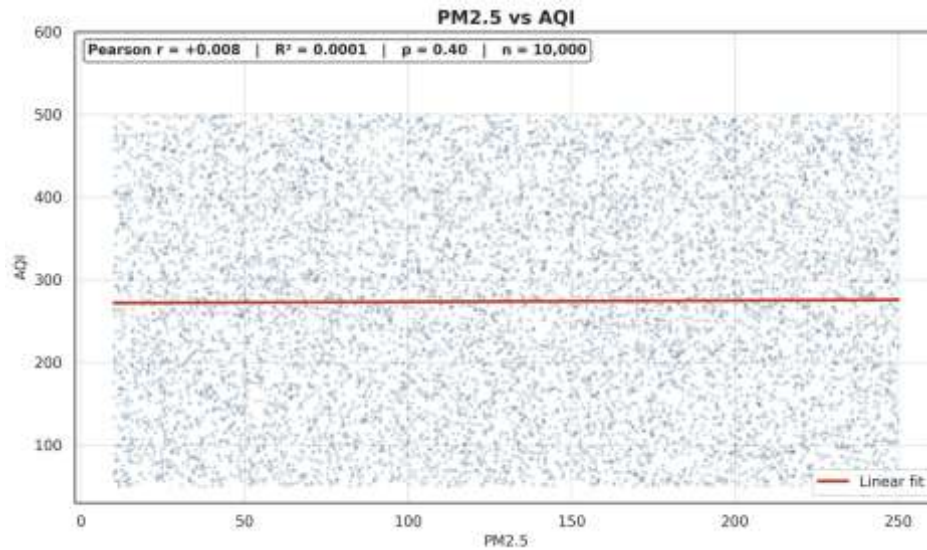
Chart: PM2.5 vs AQI

Figure 4. PM2.5 versus AQI with linear fit and correlation statistics

What the chart shows	Each of the 10,000 observations plotted as PM2.5 (x) against AQI (y), with a linear fit and printed Pearson r , R^2 and p -value.
Key insight	The fitted line is flat: $r = +0.008$, $R^2 = 0.00007$ (0.007% of variance explained), $p = 0.40$. Spearman $\rho = +0.008$ agrees.
Business / real-world implication	AQI is calculated from pollutant concentrations, with fine particulates a major component, so a clear positive relationship is expected. Its absence means either the AQI values were not derived from these pollutant columns or the values are not real measurements.
Analyst takeaway	PM2.5 explains effectively none of the variation in AQI. Not usable as a driver in this dataset.

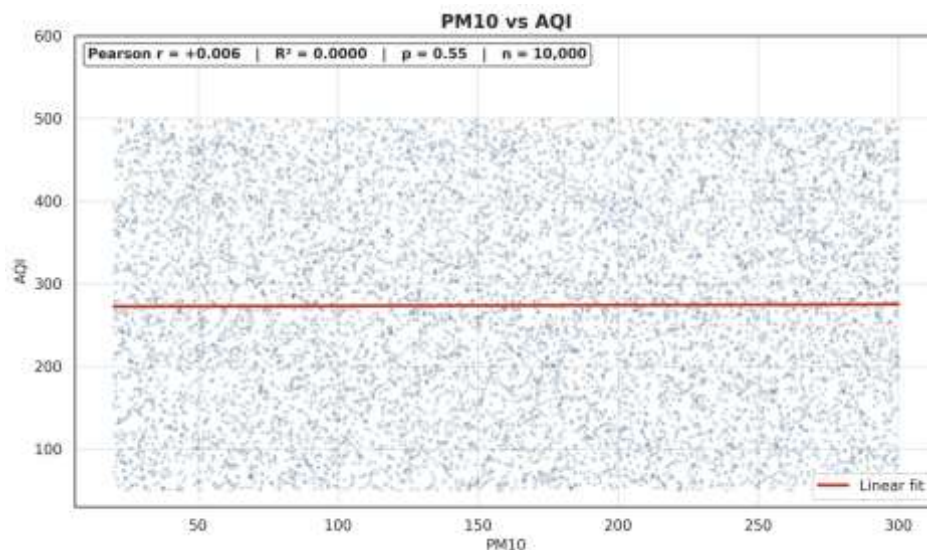
Chart: PM10 vs AQI

Figure 5. PM10 versus AQI with linear fit and correlation statistics

What the chart shows	Each of the 10,000 observations plotted as PM10 (x) against AQI (y), with a linear fit and printed Pearson r , R^2 and p -value.
Key insight	The fitted line is flat: $r = +0.006$, $R^2 = 0.00004$ (0.004% of variance explained), $p = 0.55$. Spearman $\rho = +0.006$ agrees.

Business / real-world implication	Coarse particulates are another core input to AQI. As with PM2.5, no relationship appears, which reinforces the concern about how AQI was generated. PM2.5 and PM10 are also nearly uncorrelated with each other ($r = +0.02$), whereas real measurements of the two normally move together.
Analyst takeaway	PM10 carries no signal about AQI, and its independence from PM2.5 is itself a data-quality warning.

Chart: Vehicle Count vs AQI

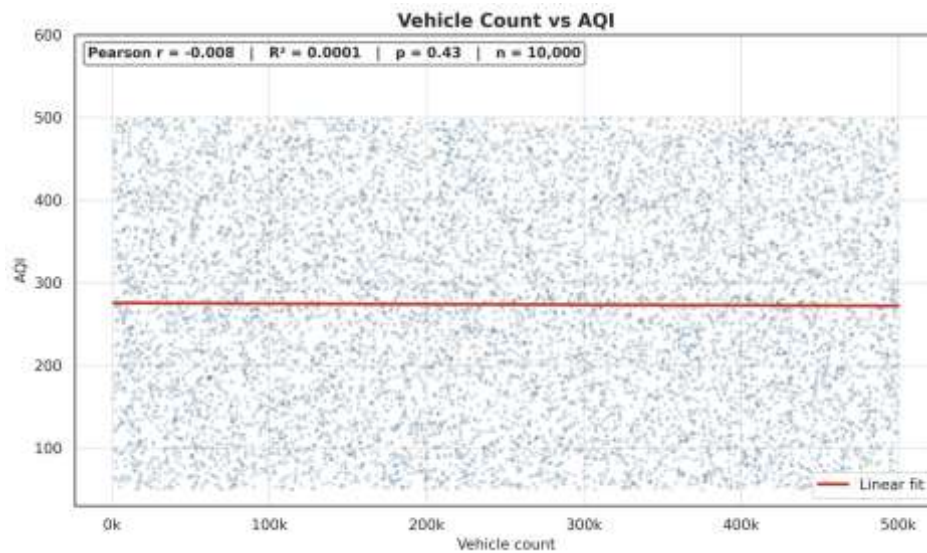


Figure 6. Vehicle count versus AQI with linear fit and correlation statistics

What the chart shows	Each of the 10,000 observations plotted as vehicle count (x) against AQI (y), with a linear fit and printed Pearson r , R^2 and p -value.
Key insight	The fitted line is flat: $r = -0.008$, $R^2 = 0.00006$ (0.006% of variance explained), $p = 0.43$. Spearman $\rho = -0.008$ agrees.
Business / real-world implication	Traffic is often proposed as a pollution driver and a target for policy. The data gives no support: the slope is marginally negative and not distinguishable from zero. This does not show that traffic is unimportant in reality, only that this dataset cannot be used to argue it either way.
Analyst takeaway	No evidence that vehicle count relates to AQI here. Do not use these data to support traffic-restriction arguments.

Chart: Industrial Activity Index vs AQI

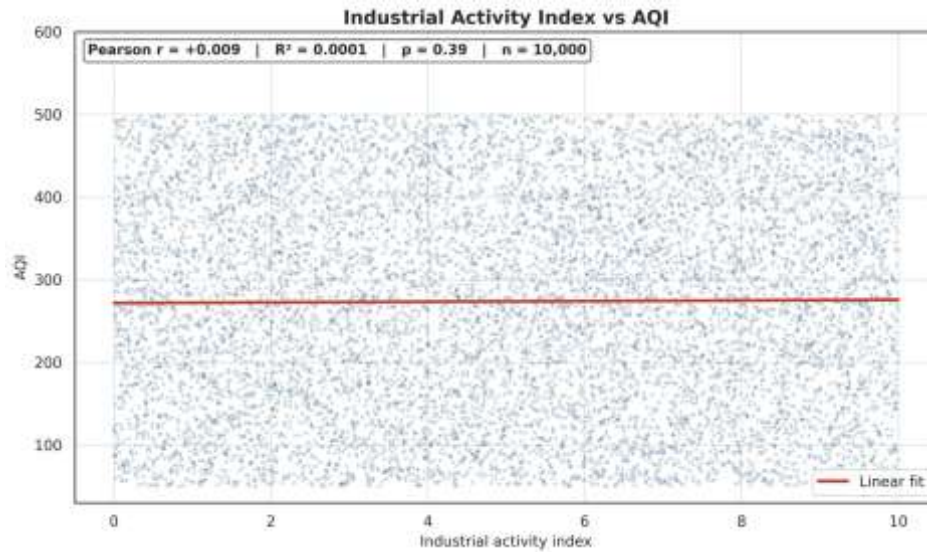


Figure 7. Industrial activity index versus AQI with linear fit and correlation statistics

What the chart shows	Each of the 10,000 observations plotted as industrial activity index (x) against AQI (y), with a linear fit and printed Pearson r, R ² and p-value.
Key insight	The fitted line is flat: r = +0.009, R ² = 0.00008 (0.008% of variance explained), p = 0.39. Spearman ρ = +0.009 agrees.
Business / real-world implication	As with traffic, the dataset cannot support a claim about industrial impact on AQI, whether positive or negative.
Analyst takeaway	No detectable link between industrial activity and AQI in this sample.

Chart: Correlation Heatmap

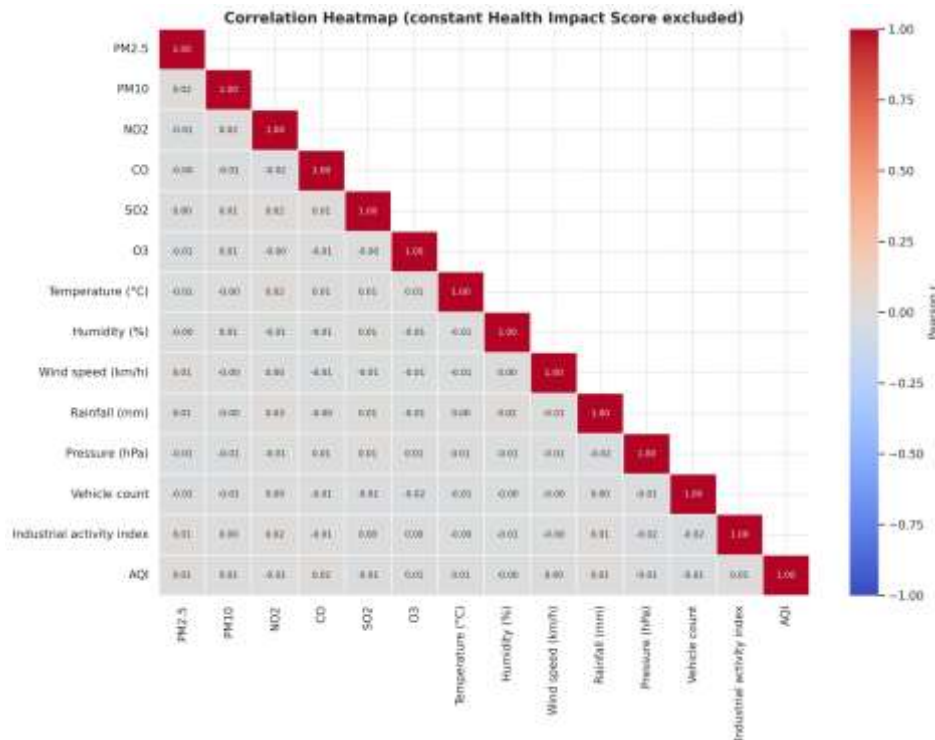


Figure 8. Pairwise Pearson correlations among AQI and the 13 explanatory variables

What the chart shows	The lower triangle of the correlation matrix for AQI and 13 variables (91 unique pairs). Health Impact Score is excluded because a constant column has undefined correlation (it appeared as a blank row in the original heatmap).
Key insight	The largest absolute correlation between any two variables is 0.029 (NO2 with Rainfall (mm)); the colour scale is fixed to -1 to $+1$, so the whole matrix is visibly neutral. Variables that normally move together, such as PM2.5 and PM10, show $r = +0.02$.
Business / real-world implication	There is no multicollinearity to manage, but there is also none of the structure real environmental data shows (pollutants moving together, weather affecting dispersion). A model built on these variables would have nothing to learn from.
Analyst takeaway	All variables are effectively independent of one another and of AQI.

Chart: Correlation of Each Variable with AQI

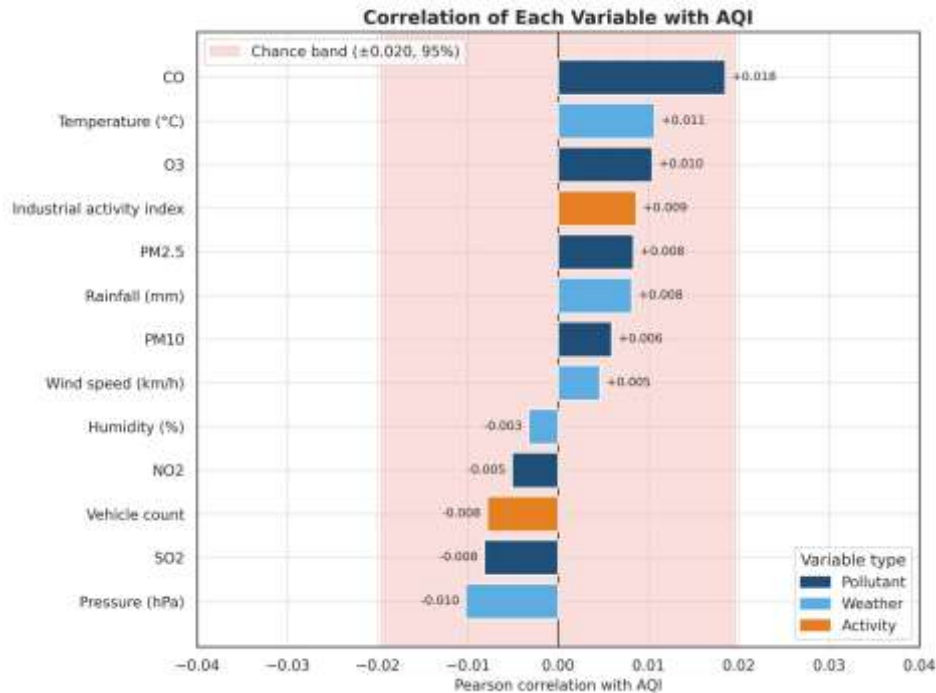


Figure 9. Pearson correlation of all 13 variables with AQI, with the 95% chance band

What the chart shows	Ranked Pearson correlations of every pollutant, weather and activity variable with AQI. The shaded band (± 0.020) marks correlations that random data of this size would produce about 95% of the time.
Key insight	The strongest association is CO at $r = +0.018$, and every bar lies inside the chance band. 0 of 13 variables are significant at 5% (about 0.7 would be expected by chance alone), and 0 after Bonferroni correction ($\alpha = 0.0038$). Rank correlations agree (max $ p = 0.019$). Combined, a linear model reaches cross-validated $R^2 = -0.004$ and a random forest -0.004 (fold range -0.008 to -0.001); negative values mean the models predict worse than simply using the average AQI.
Business / real-world implication	Ranking correlations without a significance benchmark would suggest CO is the top driver of AQI. Adding the chance band shows that would be a false positive. No pollutant, weather or activity lever can be identified from this dataset.
Analyst takeaway	The apparent “top driver” is noise. This chart is the most important guard against over-interpretation in the project.

Chart: Pollutant Box Plots

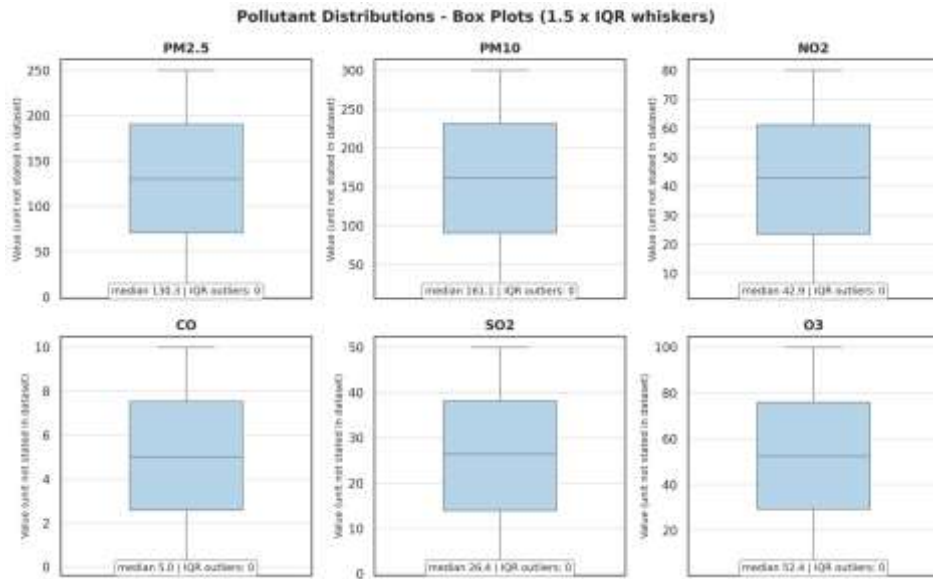


Figure 10. Box plots of the six pollutants with median and outlier count

What the chart shows	Box plot for each pollutant showing median, interquartile range and 1.5×IQR whiskers, with the number of outliers annotated. Units are not stated in the dataset.
Key insight	There are 0 outliers in any variable. Whiskers run to the same limits as the observed minimum and maximum (PM2.5 10.02–250.00, PM10 20.02–299.98, NO2 5.01–79.99, CO 0.10–10.00, SO2 2.00–49.99, O3 5.02–100.00), and medians sit near the middle of each range.
Business / real-world implication	Real pollutant series typically contain spikes far above the median, which appear as outliers. Hard, round-number ceilings and symmetric boxes indicate bounded, evenly spread values. Severe pollution episodes are therefore not represented, which limits any risk assessment.
Analyst takeaway	No outliers and round-number limits are consistent with generated rather than measured values. No outlier treatment was needed.

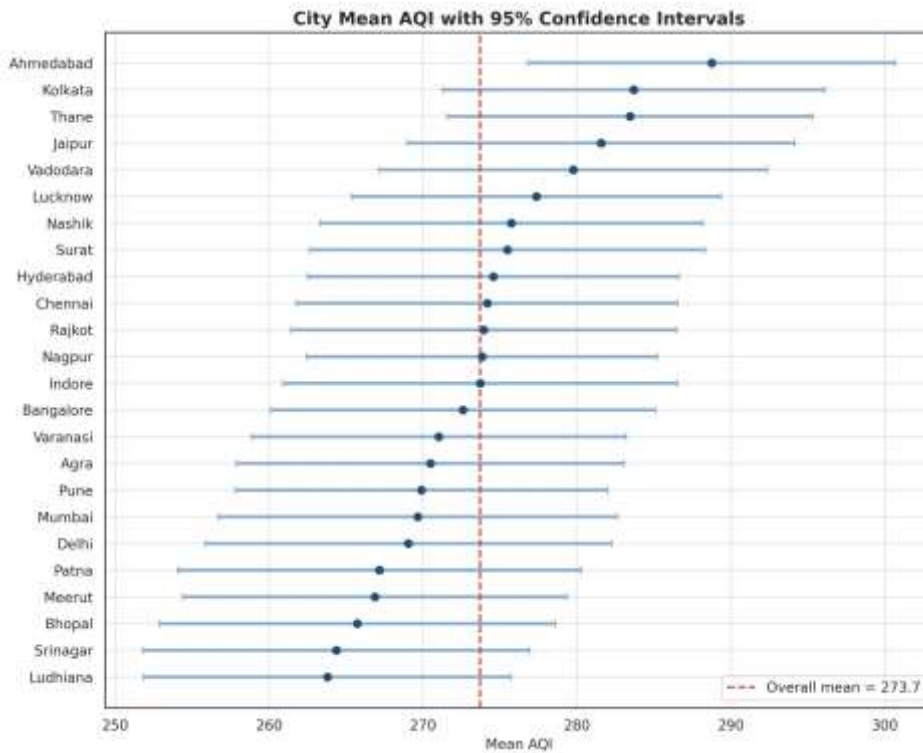
Chart: City Mean AQI with 95% Confidence Intervals

Figure 11. City mean AQI with 95% confidence intervals and the overall mean

What the chart shows	Mean AQI for each city with a 95% confidence interval, and the overall mean as a vertical reference line.
Key insight	The intervals overlap heavily: each spans about 24 points (for example Ahmedabad 276.8–300.7), roughly the same as the 25.0-point gap between the highest and lowest city means. Only Ahmedabad's interval excludes the overall mean, and after correcting for 24 comparisons its adjusted p-value is 0.28 (raw 0.012). ANOVA ($p = 0.405$) and Kruskal–Wallis ($p = 0.405$) find no overall city effect. With 24 tests at the 5% level, roughly 1.2 nominal “discoveries” are expected by chance.
Business / real-world implication	Publishing “Ahmedabad has the worst air” on this evidence would be a multiple-comparison error. Differences of this size would be expected from sampling variation alone.
Analyst takeaway	City differences are not statistically supported. The ranking in Figure 3 should be regarded as provisional at most.

Chart: Particulate Levels by AQI Category

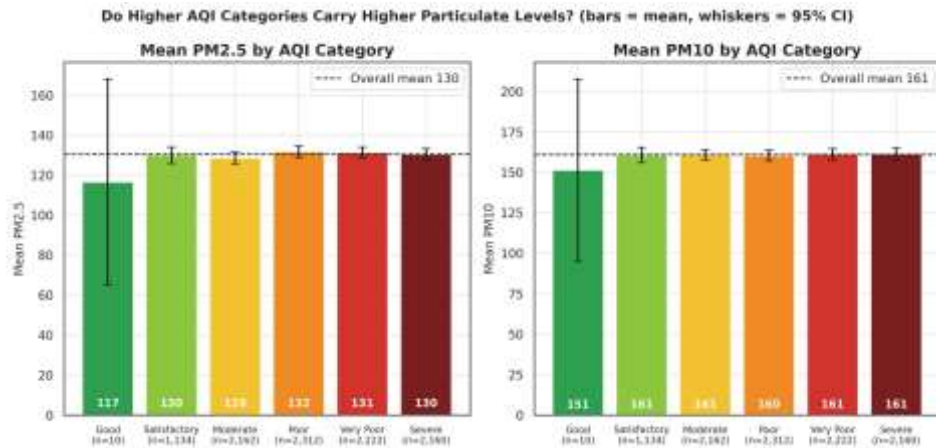


Figure 12. Mean PM2.5 and PM10 by AQI category with 95% confidence intervals

What the chart shows	Average PM2.5 and PM10 within each AQI category, with 95% confidence intervals and category sizes.
Key insight	Among the five well-populated categories, mean PM2.5 stays between 128.6 and 131.7 and mean PM10 between 160.5 and 161.4. Severe air has only 0.30 more PM2.5 than Satisfactory air ($p = 0.905$; Kruskal–Wallis across categories $p = 0.699$). The Good category has only 10 observations, hence its wide interval.
Business / real-world implication	If AQI reflected particulate pollution, Severe readings would show clearly higher PM levels than Satisfactory ones. They do not. Using reference breakpoints for all six pollutants, the highest category any row could imply is Very Poor, yet 2,160 rows (21.6%) are labelled Severe. Conversely 3,989 rows (39.9%) have a reported category below what their own pollutants imply, which the worst-sub-index rule does not allow.
Analyst takeaway	AQI categories in this file are not linked to pollutant levels: the share of rows with reported category at or above the implied one is 60.1%, the same as random shuffling (60.3%).

Chart: AQI Category Mix by City

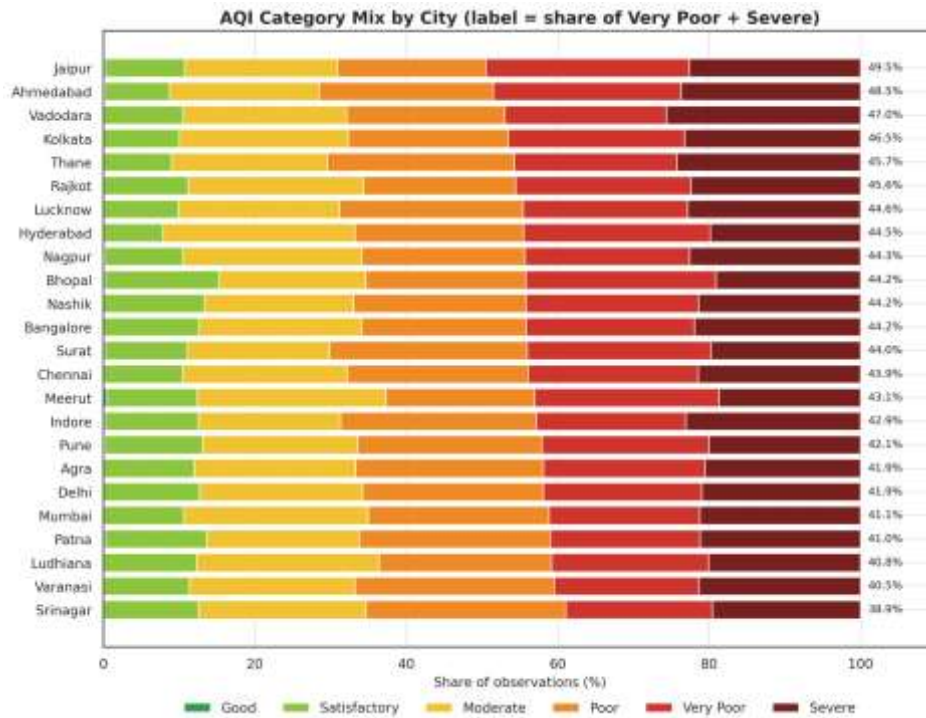


Figure 13. Share of each AQI category within each city, sorted by Very Poor + Severe share

What the chart shows	Each bar shows the percentage of a city's observations in each AQI category, sorted by the combined Very Poor and Severe share (labelled at right).
Key insight	The Very Poor + Severe share ranges from 38.9% (Srinagar) to 49.5% (Jaipur), a 10.6-point spread around the overall 43.8%. With about 417 observations per city, the standard error of such a share is roughly ± 2.4 points, so this spread is what sampling alone would produce. A chi-square test of city versus category is not significant ($p = 0.566$).
Business / real-world implication	The composition view confirms the mean-AQI result from a different angle: no city has a distinctly worse or better category profile. A decision-maker would find little to separate cities in this dataset.
Analyst takeaway	Category composition is similar in every city and does not support geographic prioritisation.

10. Key Findings

These are data-driven statements only; interpretation follows in Section 11.

1. Mean AQI is 273.7 (median 273, range 50–499); 66.9% of observations are Poor or worse, 21.6% are Severe, 0.1% are Good.
2. Ahmedabad has the highest mean AQI (288.8) and Ludhiana the lowest (263.8); the range is 25.0 points (9.1% of the overall mean).
3. City differences are not statistically significant: ANOVA $p = 0.405$; no city is significant after Bonferroni correction (Ahmedabad: raw $p = 0.012$, adjusted $p = 0.28$).
4. No pollutant, weather or activity variable is significantly correlated with AQI ($\max |r| = 0.018$); models using all 13 together score $R^2 \approx 0$ or below in cross-validation.
5. PM2.5 and PM10 are almost uncorrelated with each other ($r = +0.02$) and with AQI ($r = +0.008$ and $+0.006$).

6. All 14 non-constant variables are consistent with a flat distribution, have hard round-number limits and contain no IQR outliers.
7. The reported AQI category is below the category implied by the recorded pollutants in 3,989 rows (39.9%), which the worst-sub-index rule does not allow; 2,160 rows (21.6%) are Severe although no pollutant reading implies more than Very Poor. Agreement is no better than random shuffling (60.1% vs 60.3%).
8. Health Impact Score is 10 in every row, and the dataset has no date column, so health-impact modelling and trend or seasonality analysis are not possible.

11. Business / Real-World Insights

These are interpretations drawn from the findings above; they are judgments, not additional data.

- **The dataset behaves like independent random draws, not monitored readings.** Flat distributions, near-zero correlations between variables that should be linked, round-number limits, no outliers, and AQI categories unrelated to pollutant levels all point the same way. This is an inference; the file's provenance is not documented.
- **Decision use is not supported.** City rankings, traffic or industry arguments and health-risk statements cannot responsibly be based on this file. The one apparent pattern (a city ranking) disappears under statistical testing.
- **The analysis itself shows professional value.** A naive workflow would have reported "Ahmedabad is the most polluted city" and "CO is the top driver of AQI". Testing against a chance benchmark and correcting for multiple comparisons reveals that neither claim is reliable.
- **A real-world version of this project needs three additions:** a timestamp (for trends and seasonality), an actual health outcome (for impact analysis), and documented AQI derivation and units (for validation).

12. Recommendations

Each recommendation is tied to a specific finding.

#	Recommendation	Supported by
1	Do not use this dataset to rank cities, prioritise interventions or argue for traffic/industrial controls.	Findings 3, 4 and 6: no significant city effect, no predictor signal
2	Obtain a date or timestamp field before attempting any trend, seasonality or 2019–2024 analysis.	Finding 8: no time field despite the file name
3	Obtain a health outcome with variation (for example hospital admissions or respiratory case counts) before any health-impact modelling.	Finding 8: Health Impact Score = 10 in all rows
4	Confirm the data source and how AQI was calculated; document units and the AQI category method.	Findings 5–7: AQI unrelated to pollutants, round-number limits
5	For follow-up portfolio work, repeat this workflow on a documented real-world source (for example open national monitoring data) and compare whether the expected pollutant–AQI relationships appear.	Findings 4–7
6	Keep the validation steps as standard practice: report p-values and effect sizes, add a chance benchmark to correlation charts, and correct for multiple comparisons when ranking groups.	Charts 9 and 11

13. Limitations

- **No time dimension:** there is no date, year or month column, so genuine trends, seasonality or year-over-year change could not be calculated.
- **No health signal:** Health Impact Score is constant, so it cannot support correlation or predictive modelling.
- **Undocumented provenance:** sampling method, stations, frequency and pollutant units are not stated; AQI derivation is unknown.
- **Assumed AQI bands and breakpoints:** category bands and the six-pollutant breakpoints used in the consistency check (and their assumed units) follow the Indian AQI scheme and are external assumptions.
- **Association, not causation:** correlations measure linear or rank association only. Modelling was limited to a linear model and a random forest with fixed settings.
- **Small groups:** the Good category has only 10 observations, so its category-level statistics are imprecise.
- **Statistical caveat:** uniformity was tested with a 20-bin chi-square test; a non-rejection is consistent with, but does not prove, a uniform distribution.

14. Conclusion

The dataset is complete and well-formed: 10,000 rows, no missing or duplicate records, and valid AQI values. Descriptively, air quality in the sample is poor, with 66.9% of readings above AQI 200. Analytically, however, no city, pollutant, weather or activity variable is distinguishable from chance, and the AQI categories are unrelated to the particulate readings they should reflect.

The responsible conclusion is that this file is suitable for demonstrating a rigorous analytical workflow but not for drawing real-world conclusions about what drives urban air quality. The project shows the skills expected of a data analyst: cleaning and validation, exploratory analysis, statistical testing, clear visual communication, honest handling of a negative result, and recommendations that follow directly from the evidence.

15. Technical Implementation / Python Workflow

The analysis is a single reproducible script (aqi_analysis.py). Run it with the CSV path and an output folder; it recreates every chart, table and the results file that feeds this report.

Stage	What the script does	Output
1. Load	Read CSV; record shape and dtypes	—
2. Quality checks	Missing values, duplicates, invalid AQI, constant columns	results.json
3. Cleaning	Standardise column names (including "/"), coerce numeric types, create AQI categories	cleaned_aqi_dataset.csv
4. EDA	Distribution statistics, feature summary, IQR outliers, city summary	tables/feature_summary.csv, city_summary.csv
5. Statistics	Chi-square uniformity, Pearson/Spearman, ANOVA, Kruskal–Wallis, Welch + Bonferroni, cross-validated models, pollutant-breakpoint consistency check	tables/aqi_correlations.csv, city_vs_rest_welch.csv
6. Visualisation	13 charts at 300 dpi with a shared style	charts/01–13 *.png
7. Export	Save cleaned data and results.json	aqi_analysis_output/

Selected code

```
# Column cleaning - note the "/" replacement (fixes the wind-speed bug)
df.columns = (df.columns.str.strip().str.lower()
               .str.replace(" ", "_", regex=False).str.replace("(", "", regex=False)
               .str.replace(")", "", regex=False).str.replace("%", "pct", regex=False)
               .str.replace("°", "", regex=False).str.replace(".", "", regex=False)
               .str.replace("/", "", regex=False))

# AQI categories
df["aqi_category"] = pd.cut(df.aqi, bins=[-1, 50, 100, 200, 300, 400, 500],
                             labels=["Good", "Satisfactory", "Moderate", "Poor", "Very Poor", "Severe"])

# Correlation with significance, then a chance benchmark ( $\pm 1.96/\sqrt{n}$ )
for c in PREDICTORS:
    r_, p_ = stats.pearsonr(df[c], df.aqi)
    rho, p_s = stats.spearmanr(df[c], df.aqi)
noise_band = 1.96 / np.sqrt(len(df))

# City vs rest (Welch t-test) with Bonferroni correction
t, p = stats.ttest_ind(a, b, equal_var=False)
p_bonf = min(1.0, p * n_cities)

# Can any combination of predictors explain AQI? (5-fold CV)
rf_cv = cross_val_score(RandomForestRegressor(n_estimators=100, max_depth=8,
        min_samples_leaf=20, random_state=42), X, y, cv=KFold(5, shuffle=True, random_state=42),
        scoring="r2")
```

Workflow improvements over the first notebook

- Fixed the “/” column-name defect so wind speed is included in the analysis.
- Replaced the hard-coded Windows file path with a command-line argument.
- Added p-values, a chance band and multiple-comparison correction; the first notebook reported correlations only.
- Excluded the constant Health Impact Score from the heatmap instead of leaving a blank row.
- Added a pollutant-breakpoint consistency check, box plots, confidence-interval and composition charts to answer questions the first set of charts could not.